

WSL INSTALLATION

WSL Installation on Windows:

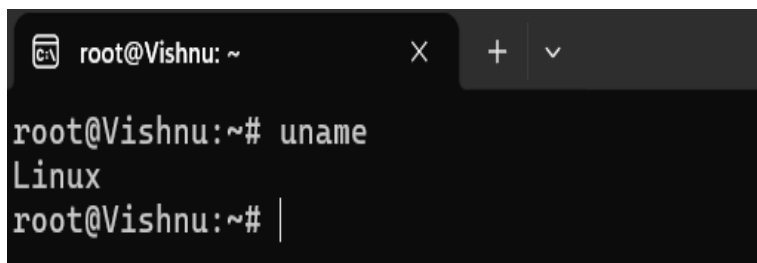
I am working on Windows, so I installed the Windows Subsystem for Linux (WSL) to get a proper Linux command-line environment instead of relying on Command Prompt or PowerShell. WSL lets Windows run a real Linux kernel and distribution (in my case, a Linux distro under WSL2) side by side with Windows, without needing a separate virtual machine or dual boot. After installing it with 'wsl --install' and setting up a Linux distribution, I get a full bash shell where standard Unix/Linux commands such as `uname`, `ls`, `cd`, `mkdir`, `ps`, `top`, `df`, and `gcc/clang` all work exactly as they would on a native Linux machine.

Below is the practical evidence of the OS-related commands executed inside my WSL terminal (running as `root@Vishnu`).

Practical Evidence

Screenshot 1: Checking the operating system using `uname`

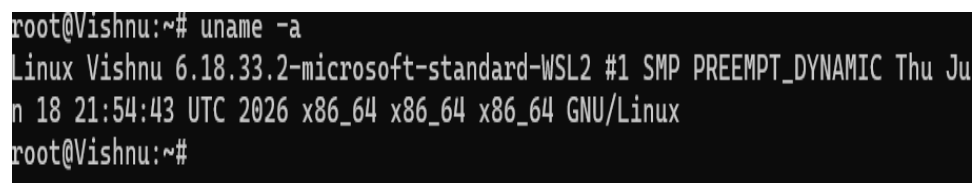
1. `uname`

A screenshot of a terminal window titled 'root@Vishnu: ~'. The terminal shows the command 'uname' being entered and executed. The output is 'Linux'. The prompt returns to 'root@Vishnu:~# |'.

Screenshot 2: Displaying detailed system information

2. `uname -a`

This shows the kernel name, hostname, kernel release/version, and machine hardware — here it confirms the Linux kernel version running under WSL2 (microsoft-standard-WSL2), along with the build date and `x86_64` architecture.

A screenshot of a terminal window showing the command 'uname -a' being executed. The output is: 'Linux Vishnu 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux'. The prompt returns to 'root@Vishnu:~#'.

Screenshot 3: Checking process information

3. `ps`

The `ps` command lists the currently running processes for the shell session, showing PID, TTY, CPU time, and the command name.

```

root@Vishnu:~# ps
  PID TTY          TIME CMD
  308 pts/0        00:00:00 bash
  459 pts/0        00:00:00 ps
root@Vishnu:~# |

```

Screenshot 4: Checking system resource usage

4. top

top gives a live view of system resource usage — CPU load, memory and swap usage, and a ranked list of running processes with their PID, owning user, and resource consumption.

```

root@Vishnu:~# top
top - 14:05:32 up 13 min,  1 user,  load average: 0.04, 0.04, 0.06
Tasks: 22 total,  1 running, 21 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.
MiB Mem :  2874.6 total,  2378.1 free,   396.5 used,   172.3 buff/cache
MiB Swap:  1024.0 total,  1024.0 free,    0.0 used.  2478.1 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
  1 root        20   0   21648   13192   9732 S   0.3   0.4   0:05.05
  2 root        20   0    3180    2200    2072 S   0.0   0.1   0:00.09
  6 root        20   0    3196    2132    2008 S   0.0   0.1   0:00.02
 61 root        19  -1   34056   12572   11500 S   0.0   0.4   0:00.86
120 root        20   0   25548    6940    5152 S   0.0   0.2   0:00.46
123 systemd+    20   0   21344   13236   11108 S   0.0   0.4   0:00.42
126 systemd+    20   0   91036    7940    6976 S   0.0   0.3   0:00.46
178 root        20   0    4244    2684    2428 S   0.0   0.1   0:00.11
179 message+    20   0    9588    5316    4636 S   0.0   0.2   0:00.44
189 root        20   0   17972    8888    7848 S   0.0   0.3   0:00.46
202 root        20   0  1755848   12924   10884 S   0.0   0.4   0:00.33
206 syslog      20   0   222516    5972    4568 S   0.0   0.2   0:00.24
209 root        20   0    3124    1972    1836 S   0.0   0.1   0:00.04
269 root        20   0  107032   23296   13844 S   0.0   0.8   0:00.56
299 root        20   0    3184    1104     980 S   0.0   0.0   0:00.00
300 root        20   0    3200    1240    1108 S   0.0   0.0   0:00.01
308 root        20   0    6076    5356    3664 S   0.0   0.2   0:00.21
314 root        20   0    6672    4496    3728 S   0.0   0.2   0:00.05
379 root        20   0   20112   11256    9320 S   0.0   0.4   0:00.37
380 root        20   0   21168    3496    1784 S   0.0   0.1   0:00.00
406 root        20   0    6080    5340    3708 S   0.0   0.2   0:00.09
460 root        20   0    9304    5700    3512 R   0.0   0.2   0:00.02

```

Conclusion

WSL successfully provides a native Linux command-line environment on top of Windows. All the required OS-level practical commands (uname, uname -a, ps, top, and similarly ls, cd, mkdir, df, gcc/clang) run directly inside the WSL terminal exactly as they would on a standalone Linux system. The screenshots above, captured from my own WSL session (root@Vishnu), serve as evidence of the practical work performed.